

Reachability and Capability Analysis for Manipulation Tasks

Oliver Porges, Theodoros Stouraitis, Christoph Borst, and Maximo A. Roa

Institute of Robotics and Mechatronics
German Aerospace Center (DLR)
82234 Wessling, Germany
`{firstname.lastname}@dlr.de`
<http://www.robotic.dlr.de>

Abstract. An offline analysis of the reachability of a robotic arm saves time for online queries like grasp selection or path planning. Reachability data is complemented with indices that quantify the goodness of one region in space to create a capability map, which can be computed based either on forward or inverse kinematics. This paper discusses the advantages and limitations of those methods, and proposes a hybrid method to improve the generation time while guaranteeing complete exploration of the space. The correctness of the results is studied with a prediction accuracy test. To illustrate the utility of a capability map, real-time visual information is incorporated to the map to help in the selection of grasp poses, or in path planning from an initial to a final pose.

Keywords: arm dexterity, reachability, manipulability, capability map, path planning.

1 Introduction

The ability of a mobile manipulator to grasp and manipulate objects depends on the location of the arm in the physical space. Arms with different kinematical structures provide different capabilities, and knowing the limits and capabilities of the arm allows a proper location of the robot to successfully fulfill a desired manipulation task. Either if the mobile robot is located in a suitable position or if the robot arm has a fixed base, an offline analysis of the robot's workspace is helpful to speed up the online solution of planning tasks. The representation of the regions where the robot tool frame (TCP) can be moved to is known as a reachability map [3]. It is in general computed as a spatial grid in the 6D space (position and orientation), where each cell has a binary value that indicates if it is reachable or not. The cells can also have an associated quality index that measures the dexterity of the robot when located in this position, thus creating a capability map (Fig. 1) [15].

An analytic representation of the reachable volumes for an arm can be obtained [7]. However, the information of the reachability map requires not only the bounding volumes, but a more complete description of the workspace, usually described with a voxelized structure. Two methods can be used to generate

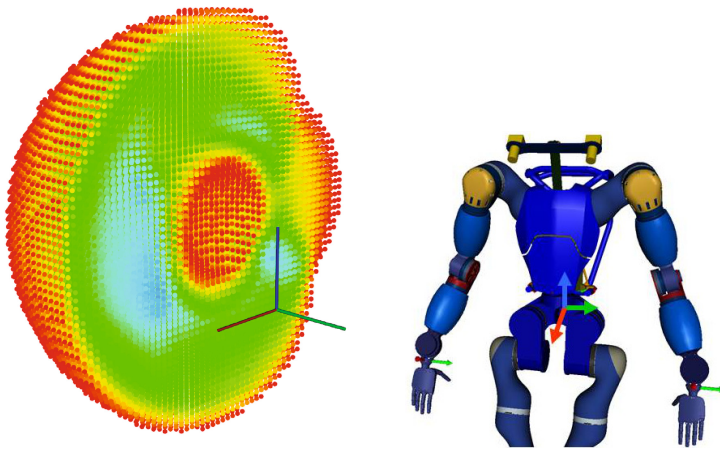


Fig. 1. Capability map for the right arm of the DLR humanoid Toro [8]

reachability maps, using either forward or inverse kinematics. With forward kinematics (FK), the arm configurations are randomly sampled in joint space, the TCP is computed and then assigned to one bin in the map, which changes value from 0 to 1. This method, however, has problems with zones close to a singularity, as too many samples tend to accumulate in a small region of the space, and it does not guarantee the completeness of the map [14]. With inverse kinematics (IK), the 6D space is uniformly sampled, and for each sample an IK solver runs to try to find a solution that leads to that configuration. This method guarantees a complete exploration of the workspace, but the use of an IK solver makes the generation of a new sample much slower [15].

The reachability or capability maps can be used for online computation and selection of reachable grasps for performing manipulation tasks [3,16]. Given a grasp pose, the position of the TCP is checked against the capability map to verify if it is reachable. To speed up query times, the reachable directions in each voxel can be represented with a simplified geometrical shape (cylinder, cone) that approximately describe the reachability properties, although at the cost of losing some precision in the analysis [16]. The map can also be applied to locate the base of a mobile manipulator for execution of 3D trajectories, like for instance opening a drawer or a door [3,5,12,14]. The associated indices that indicate the quality of a voxel help to define an optimal solution for the location of the mobile robot.

Even though the generation of maps is performed offline due to the large amount of data that must be processed, an efficient computation of the data is desirable to reduce the computational times to a minimum, while providing a data structure that facilitates online queries on the map. In cases when an FK approach is used, no criteria for stopping the sampling have been suggested, nor the validity of the map thus generated has been evaluated. This paper compares the performance of forward and inverse kinematic methods, and proposes